

01_exploratory_data_analysis

August 26, 2025

1 Customer Churn: Exploratory Data Analysis (EDA)

This notebook is part of the Telco Customer Churn machine learning project that is using the IBM dataset.

EDA phase is important for understanding and determining the patterns and properties of the dataset. This phase has a huge importance in trying to understand the data and the relations between the variables.

```
[1]: import pandas as pd
      from data_preprocessing import print_basic_info, clean_preprocess_data
      import plot_utils
```

1.1 Initial Data Exploration

```
[2]: df = pd.read_csv('../data/raw/telco-customer-churn-ibm.csv')
      df.head()
```

```
[2]:  customerID  gender  SeniorCitizen  Partner  Dependents  tenure  PhoneService  \
0  7590-VHVEG  Female                0      Yes           No         1           No
1  5575-GNVDE   Male                0      No            No        34           Yes
2  3668-QPYBK   Male                0      No            No         2           Yes
3  7795-CFOCW   Male                0      No            No        45           No
4  9237-HQITU  Female                0      No            No         2           Yes
```

```
      MultipleLines  InternetService  OnlineSecurity  ...  DeviceProtection  \
0  No phone service                DSL                No  ...                No
1                      No                DSL                Yes  ...                Yes
2                      No                DSL                Yes  ...                No
3  No phone service                DSL                Yes  ...                Yes
4                      No      Fiber optic                No  ...                No
```

```
      TechSupport  StreamingTV  StreamingMovies  ...  Contract  PaperlessBilling  \
0              No              No                No  Month-to-month                Yes
1              No              No                No      One year                No
2              No              No                No  Month-to-month                Yes
3              Yes              No                No      One year                No
4              No              No                No  Month-to-month                Yes
```

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes

[5 rows x 21 columns]

1.1.1 There are duplicate rows and some missing values that will need to be cleaned. I have also identified customerID column which is not required for our churn analysis. This column will be removed.

I will also convert **MonthlyCharges** to a numeric value behind the scene. This seems to be the only column that is supposed to be a numerical value

```
[3]: clean_preprocess_data(df)
```

```
[INFO] Missing values before: 11
[INFO] Duplicate rows before: 22
[INFO] Missing values after: 0
[INFO] Duplicate rows after: 0
[INFO] Final dataset shape: (7010, 20)
```

```
[3]:      gender  SeniorCitizen  Partner  Dependents  tenure  PhoneService  \
0    Female              0     Yes         No         1           No
1     Male              0     No         No        34           Yes
2     Male              0     No         No         2           Yes
3     Male              0     No         No        45           No
4    Female              0     No         No         2           Yes
...    ...              ...     ...         ...         ...           ...
7038   Male              0     Yes         Yes        24           Yes
7039  Female              0     Yes         Yes        72           Yes
7040  Female              0     Yes         Yes        11           No
7041   Male              1     Yes         No         4           Yes
7042   Male              0     No         No        66           Yes
```

	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	\
0	No phone service	DSL	No	Yes	
1	No	DSL	Yes	No	
2	No	DSL	Yes	Yes	
3	No phone service	DSL	Yes	No	
4	No	Fiber optic	No	No	
...	...	...	...	...	
7038	Yes	DSL	Yes	No	
7039	Yes	Fiber optic	No	Yes	
7040	No phone service	DSL	Yes	No	

7041	Yes	Fiber optic	No	No
7042	No	Fiber optic	Yes	No

	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract \
0	No	No	No	No	Month-to-month
1	Yes	No	No	No	One year
2	No	No	No	No	Month-to-month
3	Yes	Yes	No	No	One year
4	No	No	No	No	Month-to-month
...	...	...	...	...	...
7038	Yes	Yes	Yes	Yes	One year
7039	Yes	No	Yes	Yes	One year
7040	No	No	No	No	Month-to-month
7041	No	No	No	No	Month-to-month
7042	Yes	Yes	Yes	Yes	Two year

	PaperlessBilling	PaymentMethod	MonthlyCharges \
0	Yes	Electronic check	29.85
1	No	Mailed check	56.95
2	Yes	Mailed check	53.85
3	No	Bank transfer (automatic)	42.30
4	Yes	Electronic check	70.70
...	...	...	...
7038	Yes	Mailed check	84.80
7039	Yes	Credit card (automatic)	103.20
7040	Yes	Electronic check	29.60
7041	Yes	Mailed check	74.40
7042	Yes	Bank transfer (automatic)	105.65

	TotalCharges	Churn
0	29.85	No
1	1889.50	No
2	108.15	Yes
3	1840.75	No
4	151.65	Yes
...	...	...
7038	1990.50	No
7039	7362.90	No
7040	346.45	No
7041	306.60	Yes
7042	6844.50	No

[7010 rows x 20 columns]

```
[4]: df.describe()
```

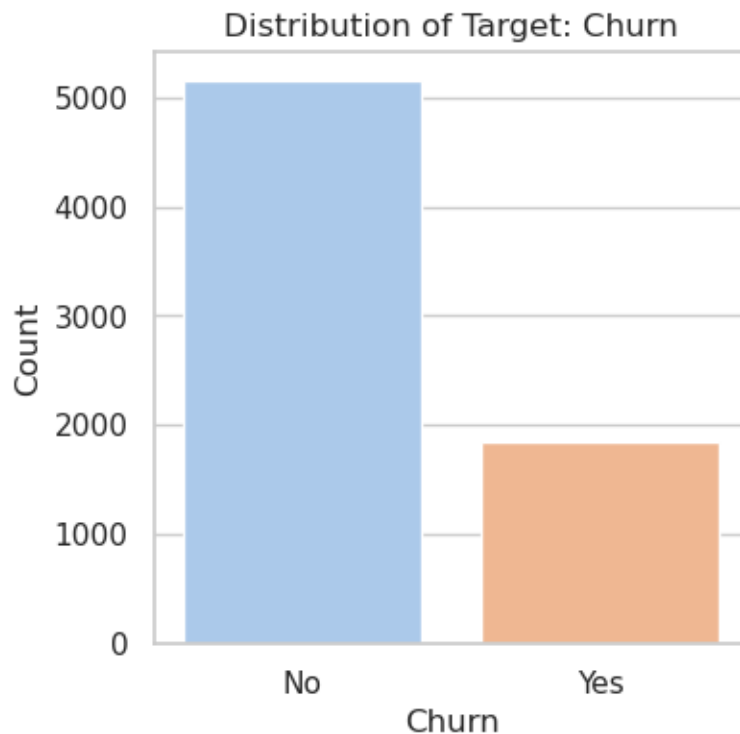
```
[4]:
```

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7010.000000	7010.000000	7010.000000	7010.000000
mean	0.162767	32.520399	64.888666	2290.353388
std	0.369180	24.520441	30.064769	2266.820832
min	0.000000	1.000000	18.250000	18.800000
25%	0.000000	9.000000	35.750000	408.312500
50%	0.000000	29.000000	70.400000	1403.875000
75%	0.000000	56.000000	89.900000	3807.837500
max	1.000000	72.000000	118.750000	8684.800000

1.1.2 Distribution of Target Variable (Churn)

- This distribution is not balanced, therefore, I will need to balance the data with SMOTE technique to ensure balanced classes before I begin with feature engineering and model training.

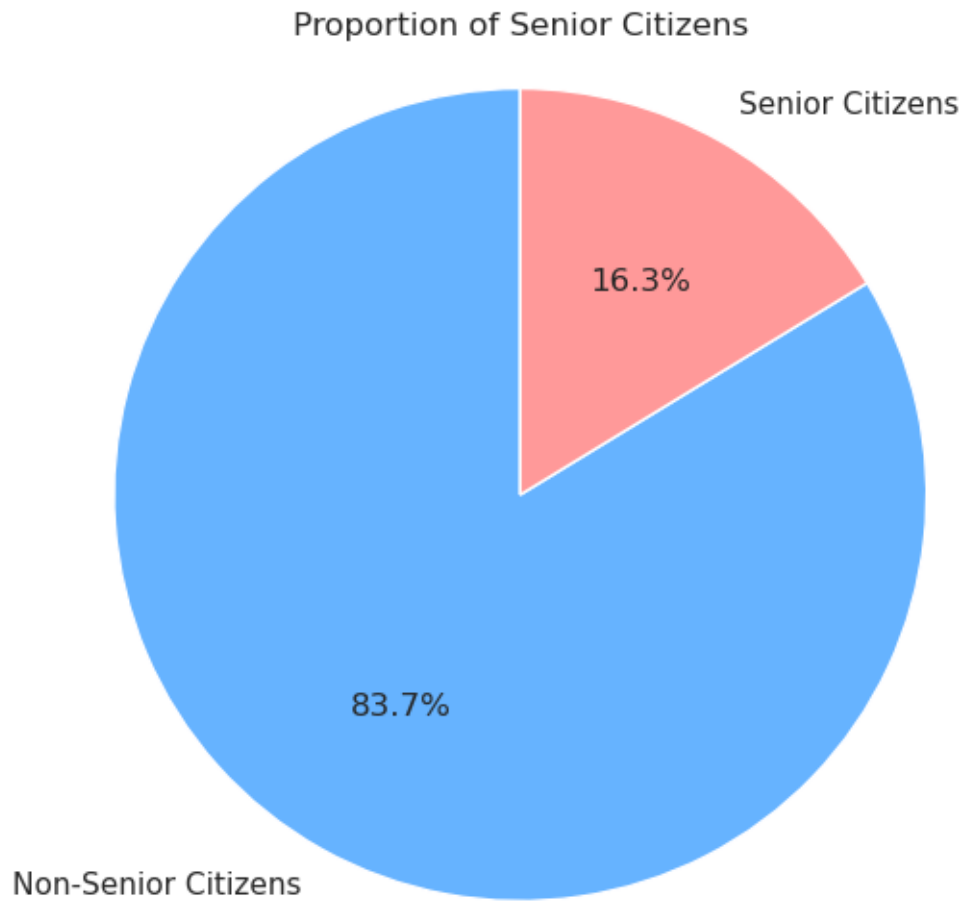
```
[5]: plot_utils.plot_target_distribution(df, target_col='Churn')
```



1.1.3 Senior Citizens in the Dataset

- Most of the customers are not senior citizens. Therefore, I do not expect churn to be influenced by this variable.

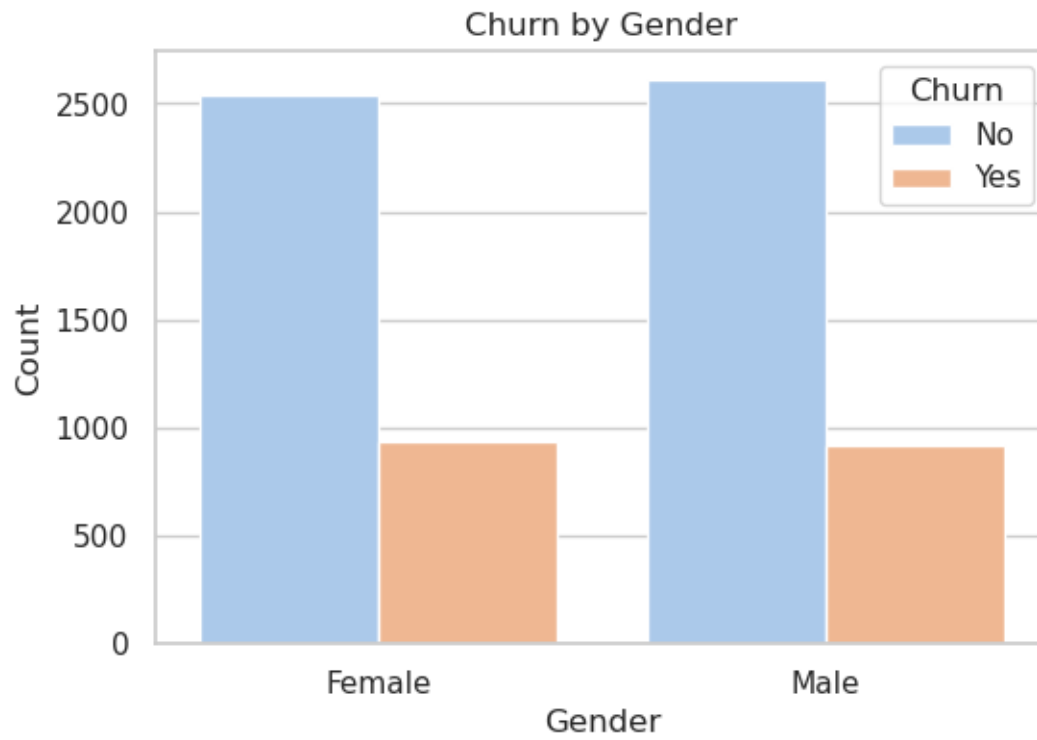
```
[6]: plot_utils.plot_senior_citizens_pie(df)
```



1.1.4 Churn by Gender

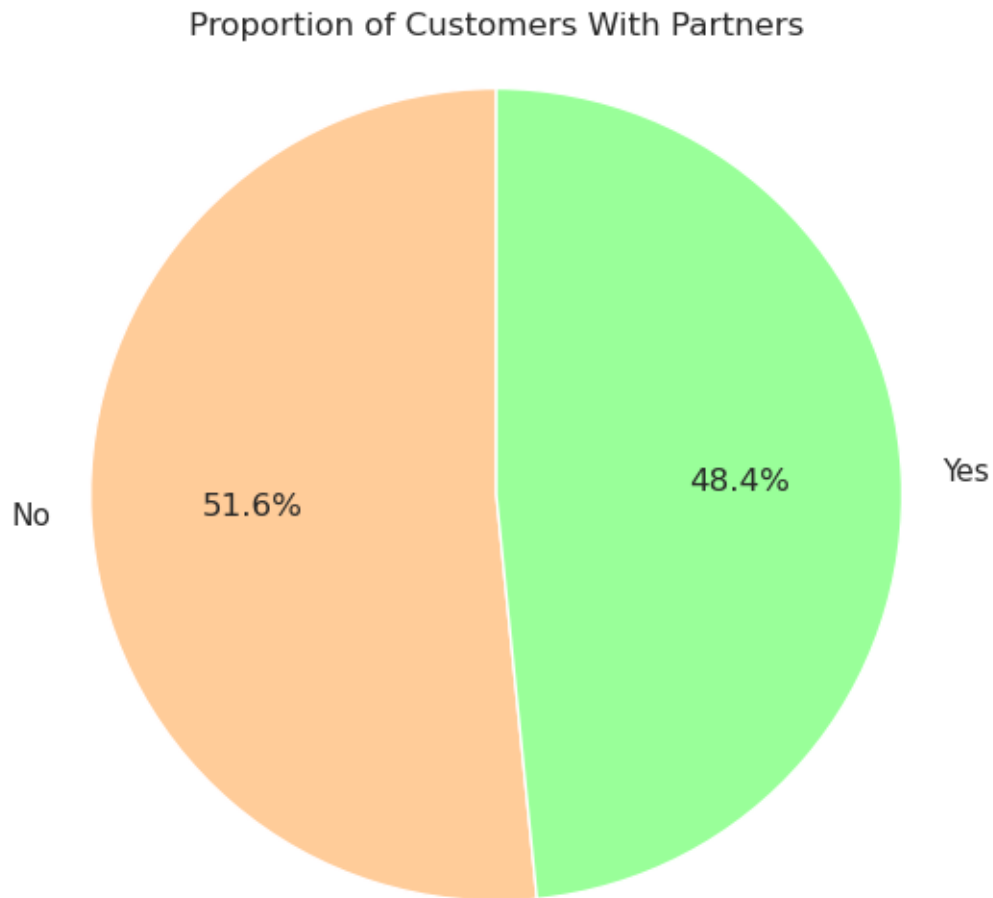
- The churn between male and female customers is almost the same, which is valuable information

```
[7]: plot_utils.plot_churn_by_gender(df)
```



1.1.5 Customers With and Without Partners

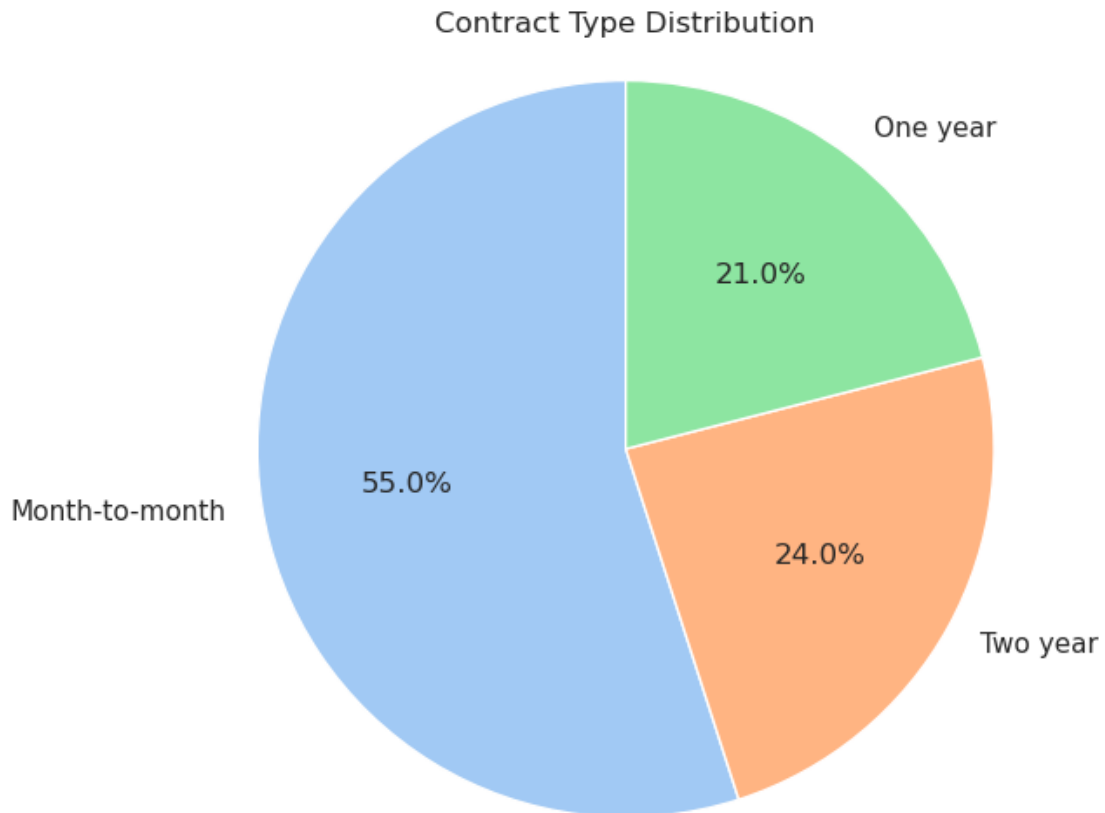
```
[8]: plot_utils.plot_partners_pie(df)
```



1.1.6 Customers by Contract Type

- More than 50% of customers are on Month-to-Month contract
- These 3 variables will be interesting to see if any of them have a high churn rate

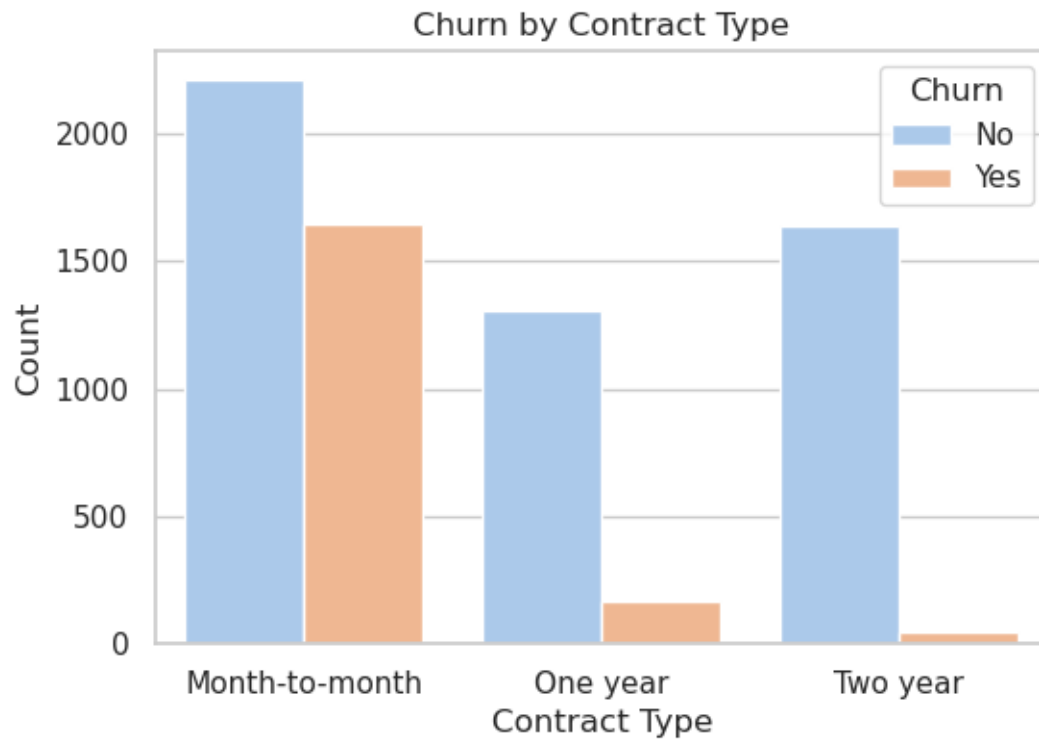
```
[9]: plot_utils.plot_contract_type_pie(df)
```



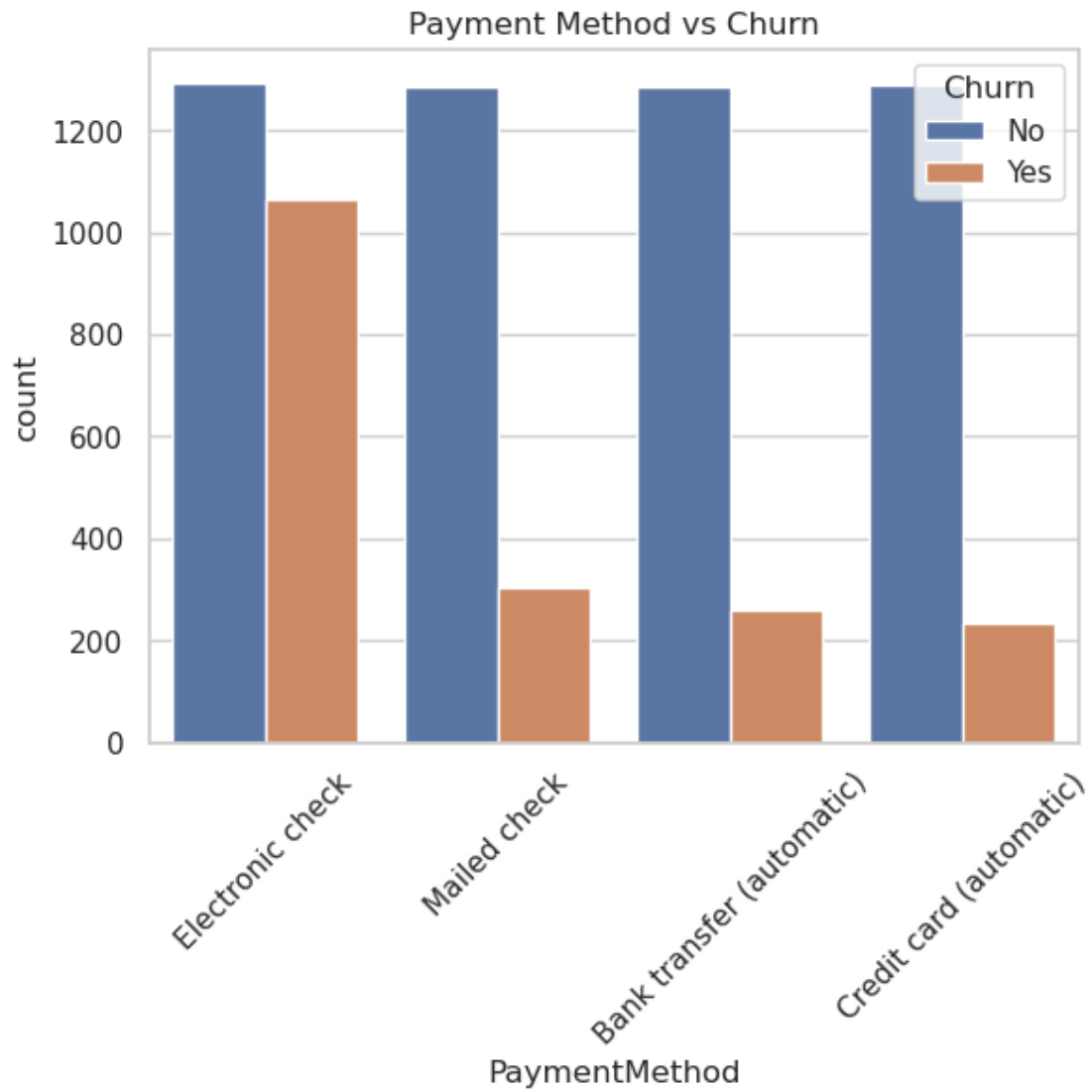
1.1.7 Churn by Contract Type

- It is clear based on the below data that most churners are on Month-to-month contract type.
- I will need to evaluate charges for each contract type.
- Could it be Month-to-month is high?
- what are the incentives of being on other contract types that are not on Month-to-month?

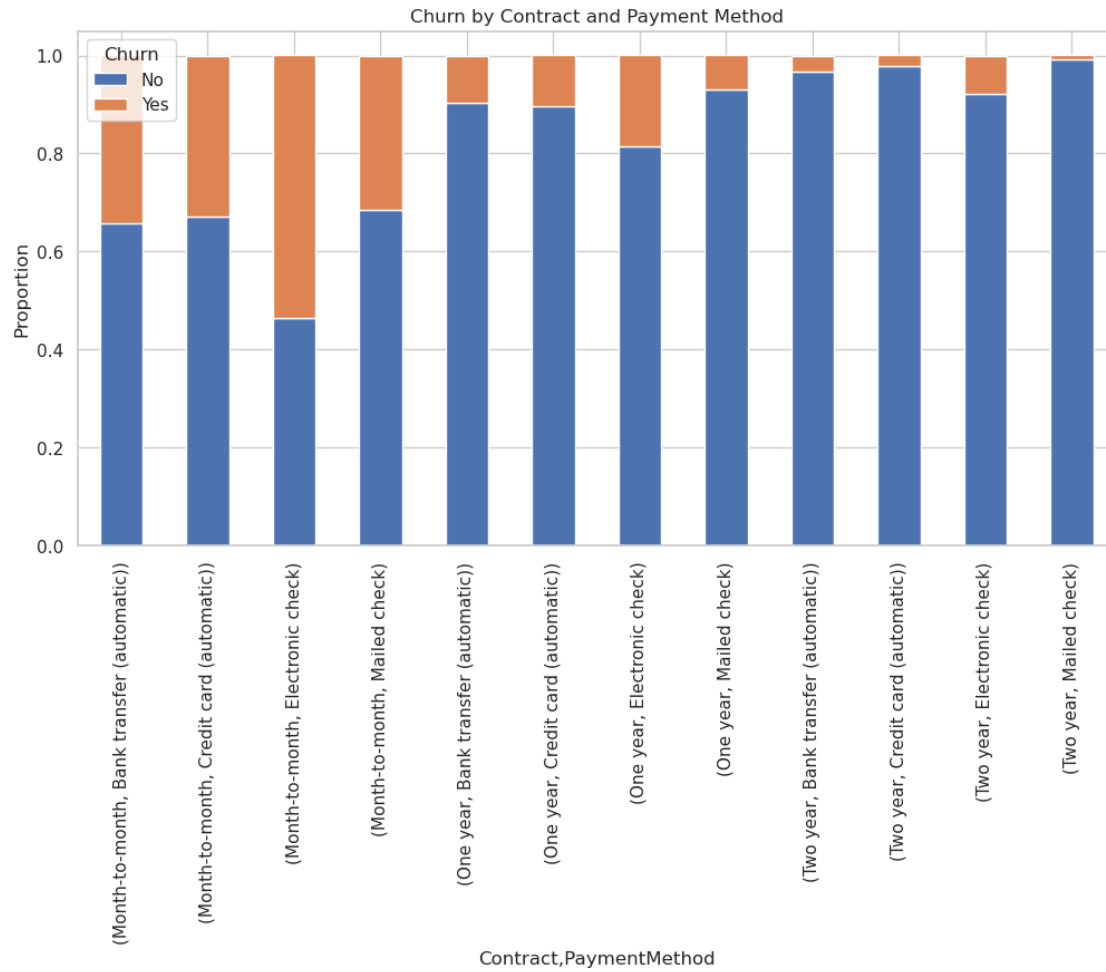
```
[10]: plot_utils.plot_churn_by_contract_type(df)
```

```
[11]: plot_utils.plot_payment_method(df)
```

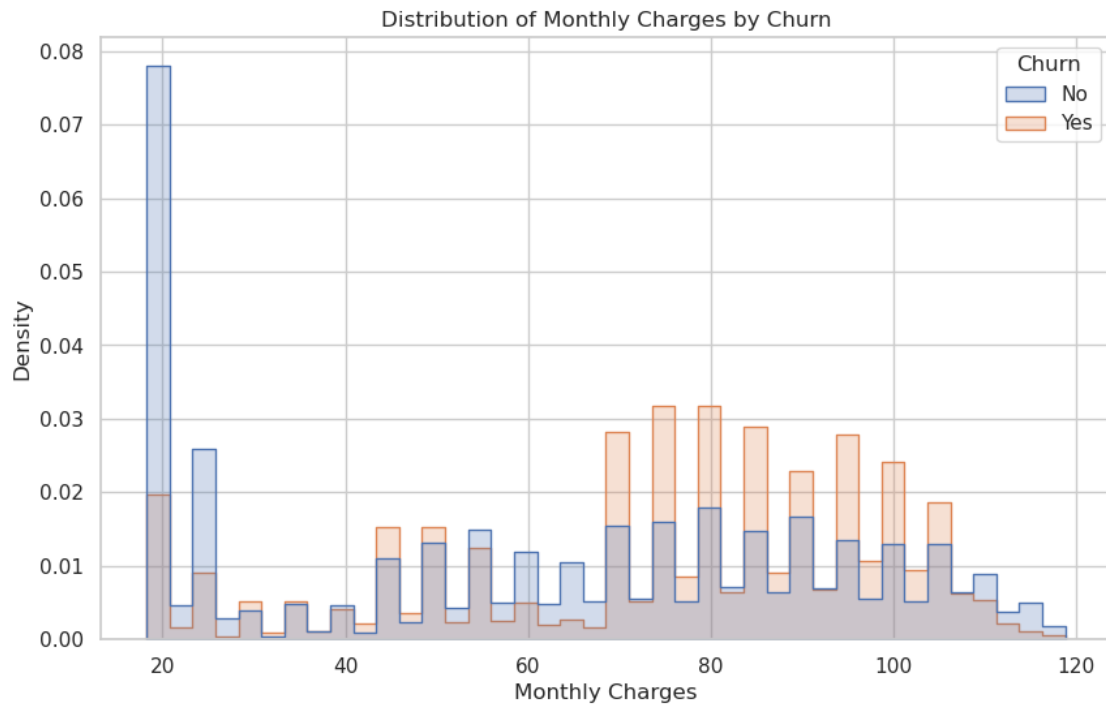


```
[12]: plot_utils.plot_contract_payment_method(df)
```

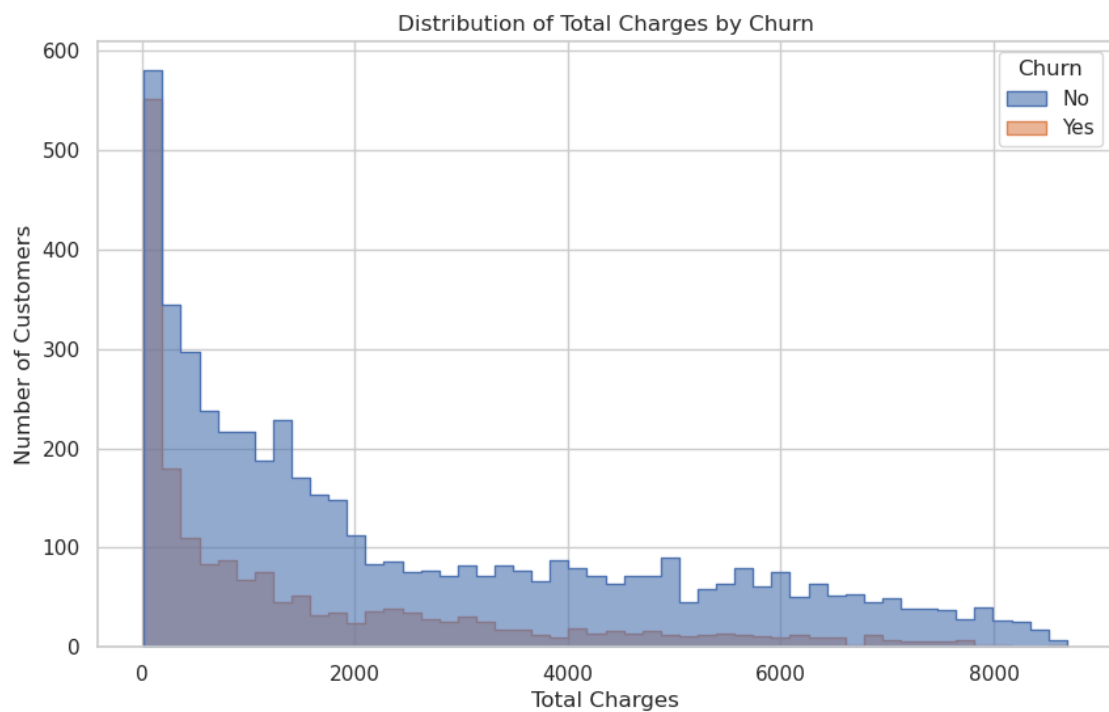


2

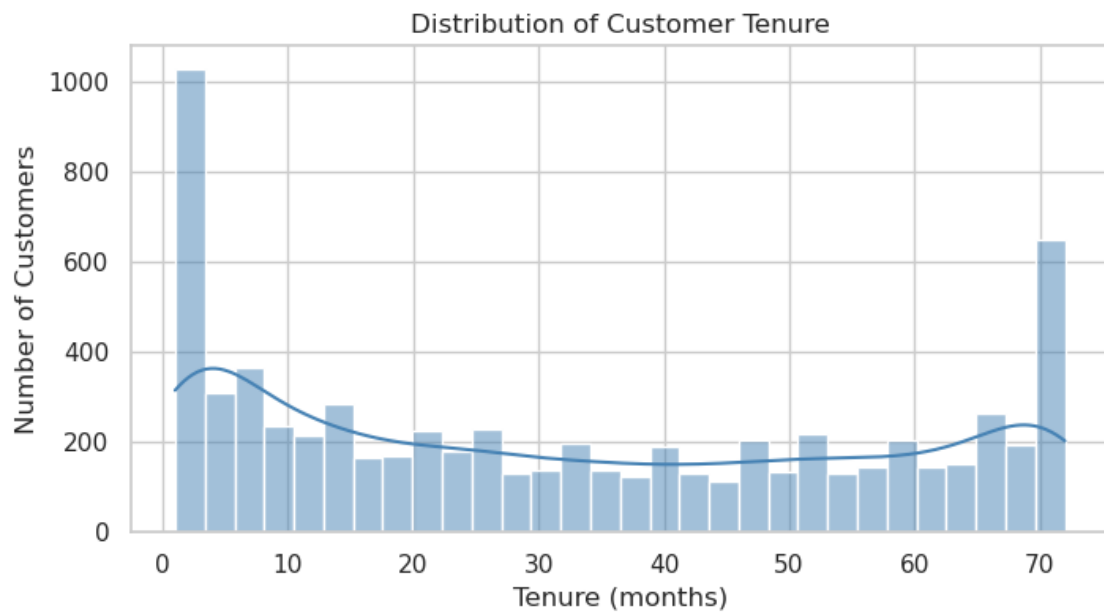
```
[13]: plot_utils.plot_monthly_charges(df)
```



```
[14]: plot_utils.plot_total_charges(df)
```

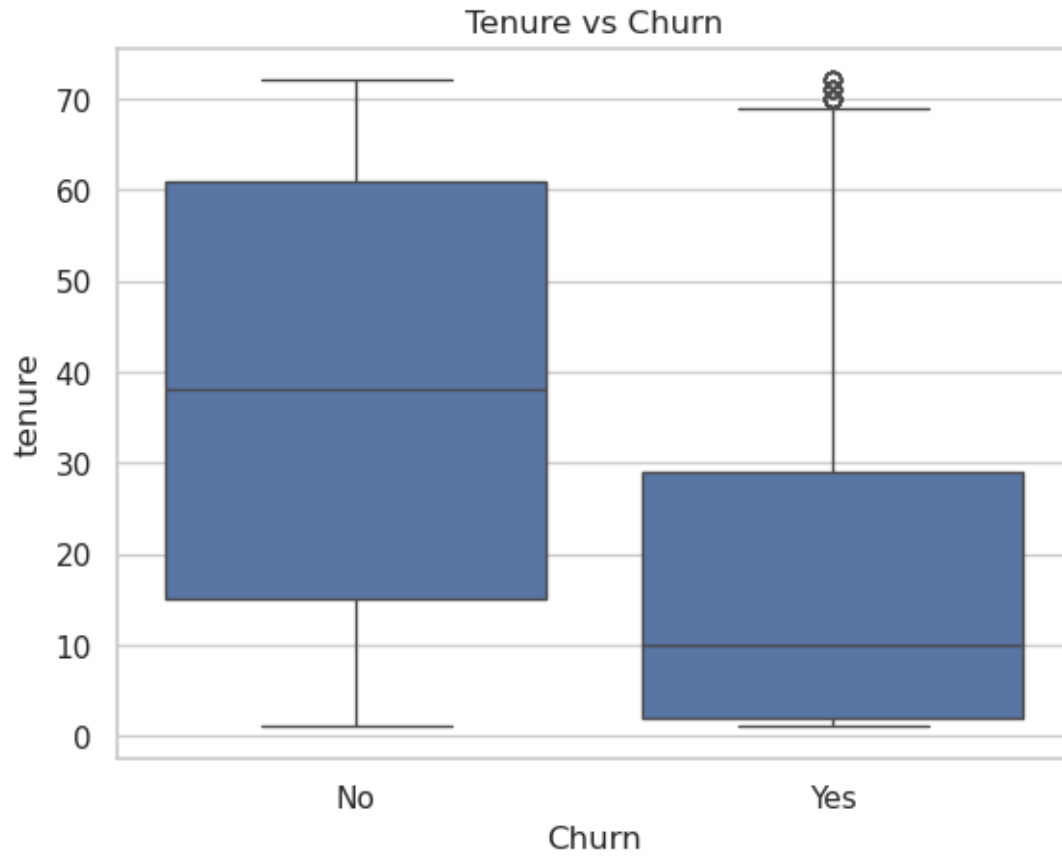


```
[15]: plot_utils.plot_tenure_distribution(df)
```



2.1 Compare distributions for churned vs non-churned customers

```
[16]: plot_utils.plot_tenure_churn(df)
```



3 Tenure Bands

3.1 Most churn happens within the first 12 months, and there it remains almost the same rate after 12 months.

3.2 Furthermore, after 48 months Churn rate is very low compared to non-churners.

```
[17]: plot_tenure_bands(df)
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[17], line 1  
----> 1 plot_tenure_bands(df)  
  
NameError: name 'plot_tenure_bands' is not defined
```

3.2.1 Correlation Heatmap

```
[ ]: df['Churn_'] = df['Churn'].map({'No': 0, 'Yes': 1})  
plot_correlation_heatmap(df)
```

3.3 Summary

- Customers on flexible (month-to-month) contracts are at higher risk of churn
- I will perform statistical analysis to better understand the dataset

3.3.1 NB: This is first eda of an iterative process, more insights will be discovered in statistical analysis.

```
[ ]:
```